

Project Harbour

Retirement drawdown risk. Monte Carlo, 20,000 paths per scenario. Figures link to the other tabs.

Headline	Result
Chance of ruin, textbook (normal returns, no health shocks)	15.2%
Chance of ruin, regimes and health shocks (base case)	24.9%
Chance of ruin, base case plus bear market in year one	40.0%
Gap between textbook and stressed case (points)	+24.8%
Chance of ruin with guardrail spending	11.9%
Chance of ruin with VPW	11.8%
Median ending wealth, base case	\$773,259
Expected ending wealth, base case	\$1,360,412
Lowest ruin across stock weights, static spending	24.8%
Stock weight at that lowest ruin	70%
Lowest ruin across stock weights, guardrails	10.7%
Stock weight at that lowest ruin, guardrails	40%
Ruin gap, worst to best fifth of opening years (points)	+61.7%
Cost of tax in ruin terms, pro rata order (points)	+2.1%
Cost of tax in ruin terms, fill low brackets (points)	+4.3%

Ruin means the money ran out and a bill went unpaid. Results are model output under stated assumptions, not for

recasts.

Stress ladder

Static spending, 60/40, \$40,000 a year. One problem added at each step. 20,000 paths.

Scenario	Chance of ruin	Chance of ending below start
1. Normal returns	15.2%	54.0%
2. Regime switching	19.6%	51.3%
3. Regimes + health shocks	24.9%	55.9%
4. Regimes + health + bear market in year one	40.0%	70.3%

Expected ending wealth	5th percentile	25th percentile	Median	75th percentile
\$1,331,796	\$0	\$240,883	\$872,482	\$1,873,558
\$1,504,815	\$0	\$154,061	\$947,680	\$2,195,425
\$1,360,412	\$0	\$3,648	\$773,259	\$1,983,805
\$875,891	\$0	\$0	\$273,597	\$1,238,366

95th percentile	Ruin vs textbook (points)
\$4,303,032	0.0%
\$5,016,387	+4.4%
\$4,773,183	+9.7%
\$3,629,506	+24.8%

Withdrawal rules

Regimes and health shocks on. VPW spends the pot down by design, so judge it on spending.

Scenario	Chance of ruin	Chance spending below 80%
Static	24.9%	24.6%
Guardrails	11.9%	50.8%
VPW	11.8%	43.5%
Static + year-one bear	40.0%	39.7%
Guardrails + year-one bear	20.0%	66.0%
VPW + year-one bear	14.3%	58.2%

Mean spending vs target	Median ending wealth	Expected ending wealth	Median lifetime tax
93.8%	\$773,259	\$1,360,412	\$91,691
100.0%	\$898,930	\$1,248,542	\$93,610
153.8%	\$0	\$0	\$76,584
88.4%	\$273,597	\$875,891	\$68,327
89.0%	\$643,746	\$934,946	\$76,560
130.6%	\$0	\$0	\$64,265

Ruin vs static (points)	
	0.0%
	-13.0%
	-13.1%
	0.0%
	-20.0%
	-25.7%

Withdrawal order and tax

Static spending, regimes and health shocks on. Control has no tax at all.

Order	Chance of ruin	Median ending wealth	Expected ending wealth
No tax (control)	20.6%	\$1,011,451	\$1,646,831
Pro rata	22.7%	\$870,360	\$1,461,084
Taxable first	23.6%	\$849,520	\$1,450,471
Traditional first	24.2%	\$799,645	\$1,391,594
Fill low brackets	24.9%	\$773,259	\$1,360,412

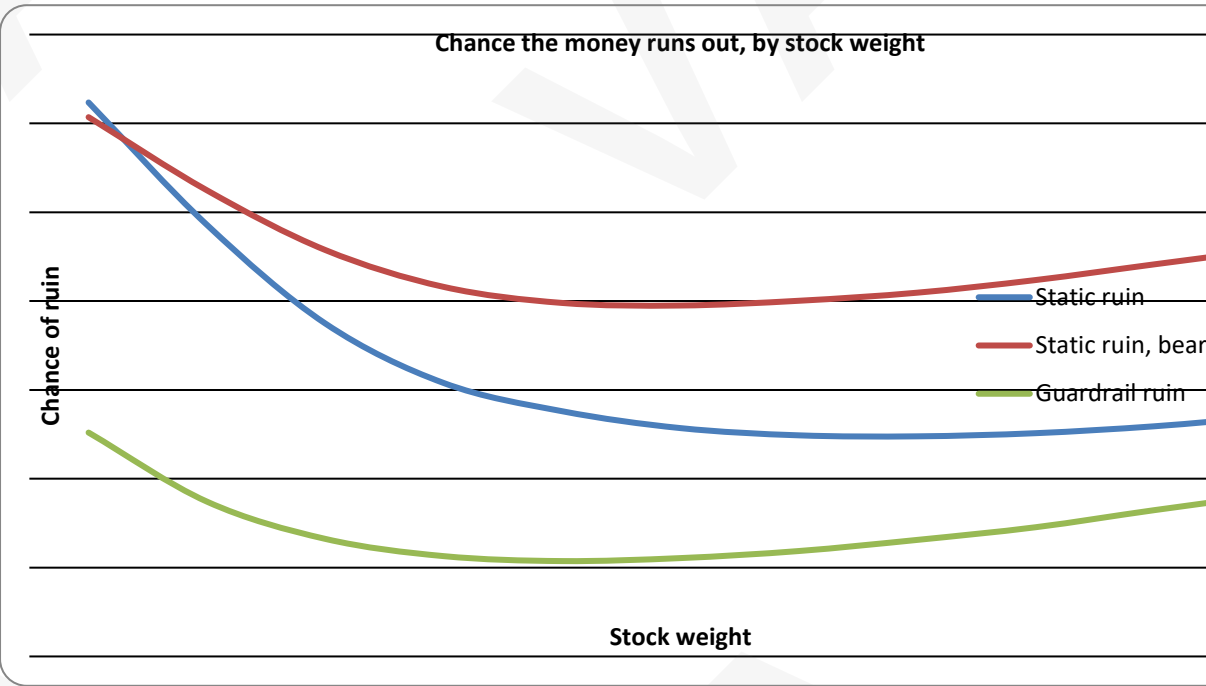
Median lifetime tax	Tax as share of withdrawals	Extra ruin vs no tax (points)	Rank by ruin (1 is best)
\$0	0.0%	0.0%	n/a
\$76,459	7.0%	+2.1%	1
\$87,850	7.7%	+3.1%	2
\$87,815	7.6%	+3.7%	3
\$91,691	7.0%	+4.3%	4

Allocation sweep

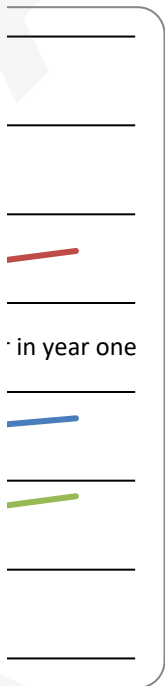
Stock weight from 0% to 100%. Regimes and health shocks on.

Stock weight	Static ruin	Static ruin, bear in year one	Guardrail ruin
0%	62.4%	60.7%	25.2%
10%	48.6%	52.5%	17.4%
20%	37.5%	45.7%	13.2%
30%	30.8%	41.6%	11.3%
40%	27.6%	39.8%	10.7%
50%	25.7%	39.5%	11.1%
60%	24.9%	40.0%	11.9%
70%	24.8%	40.9%	13.1%
80%	25.1%	42.4%	14.5%
90%	25.9%	44.1%	16.5%
100%	27.0%	45.9%	18.3%

Lowest ruin	24.8%	39.5%	10.7%
Stock weight at lowest ruin	70%	50%	40%



Static median ending wealth	Guardrail median ending wealth	Static chance pot halves	Static median worst drawdown
\$0	\$187,665	94.3%	100.0%
\$14,156	\$298,582	87.2%	98.6%
\$145,307	\$416,107	75.2%	86.2%
\$290,791	\$541,350	62.9%	73.2%
\$446,109	\$662,500	53.1%	62.1%
\$606,297	\$779,697	46.8%	55.1%
\$773,259	\$898,930	43.0%	52.5%
\$948,163	\$1,015,894	41.2%	53.2%
\$1,116,835	\$1,134,811	40.6%	54.9%
\$1,267,808	\$1,252,742	40.6%	57.8%
\$1,400,039	\$1,358,685	41.4%	60.9%

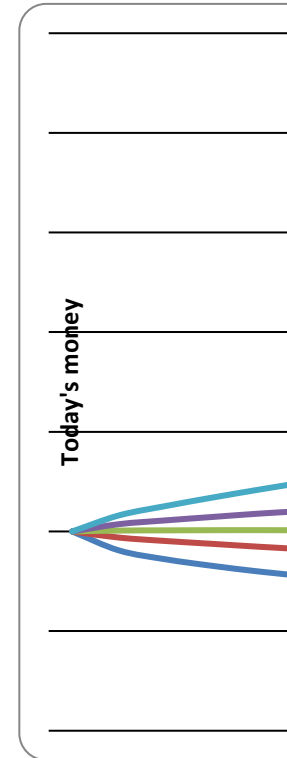


Portfolio percentiles by age

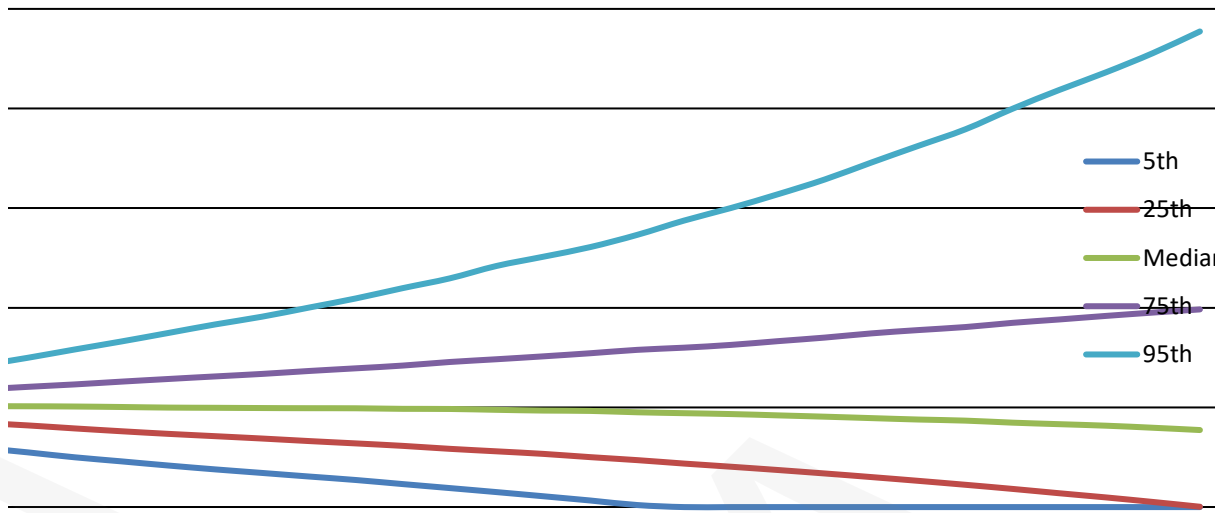
Base case. Regimes and health shocks on, static spending, 60/40.

Age	5th	10th	25th	Median	75th
65	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000
66	\$809,298	\$865,367	\$938,419	\$1,008,976	\$1,073,327
67	\$719,877	\$795,505	\$903,937	\$1,010,970	\$1,111,507
68	\$654,139	\$740,506	\$875,251	\$1,012,606	\$1,145,596
69	\$597,208	\$687,122	\$845,987	\$1,012,791	\$1,179,339
70	\$549,346	\$645,163	\$819,756	\$1,011,703	\$1,206,603
71	\$499,916	\$602,671	\$789,622	\$1,009,419	\$1,231,162
72	\$458,951	\$561,982	\$761,279	\$1,004,466	\$1,259,082
73	\$417,099	\$523,930	\$735,079	\$998,738	\$1,285,736
74	\$378,461	\$491,954	\$711,694	\$996,516	\$1,311,455
75	\$342,745	\$459,536	\$688,057	\$994,069	\$1,336,474
76	\$306,825	\$424,937	\$662,681	\$992,395	\$1,366,094
77	\$271,717	\$386,683	\$639,032	\$991,634	\$1,392,648
78	\$230,934	\$351,917	\$614,396	\$986,051	\$1,420,244
79	\$190,952	\$314,471	\$584,872	\$984,674	\$1,456,182
80	\$150,494	\$274,757	\$559,667	\$976,173	\$1,483,668
81	\$108,746	\$235,116	\$533,793	\$968,642	\$1,513,095
82	\$66,097	\$196,775	\$501,325	\$965,195	\$1,543,934
83	\$20,666	\$156,670	\$471,301	\$951,560	\$1,578,079
84	\$0	\$118,143	\$436,837	\$942,481	\$1,600,108
85	\$0	\$75,417	\$405,220	\$933,656	\$1,627,810
86	\$0	\$31,325	\$372,137	\$920,577	\$1,665,236
87	\$0	\$0	\$337,398	\$907,370	\$1,701,253
88	\$0	\$0	\$301,092	\$892,984	\$1,743,591
89	\$0	\$0	\$263,217	\$878,731	\$1,776,781
90	\$0	\$0	\$223,884	\$867,852	\$1,807,031
91	\$0	\$0	\$182,776	\$847,430	\$1,848,872
92	\$0	\$0	\$138,901	\$832,132	\$1,883,021
93	\$0	\$0	\$96,951	\$816,905	\$1,918,799
94	\$0	\$0	\$51,420	\$796,326	\$1,952,194
95	\$0	\$0	\$3,648	\$773,259	\$1,983,805

90th	95th	Mean	Share still solvent
\$1,000,000	\$1,000,000	\$1,000,000	100.0%
\$1,129,179	\$1,161,110	\$1,001,207	100.0%
\$1,200,619	\$1,255,523	\$1,003,828	100.0%
\$1,267,303	\$1,340,463	\$1,008,141	100.0%
\$1,329,709	\$1,420,877	\$1,012,999	100.0%
\$1,387,825	\$1,498,119	\$1,016,499	100.0%
\$1,442,100	\$1,581,462	\$1,019,120	100.0%
\$1,503,972	\$1,662,354	\$1,024,350	100.0%
\$1,562,524	\$1,747,604	\$1,028,834	100.0%
\$1,625,798	\$1,833,201	\$1,035,255	100.0%
\$1,689,455	\$1,910,996	\$1,043,207	100.0%
\$1,747,053	\$2,002,380	\$1,049,720	99.9%
\$1,810,988	\$2,094,297	\$1,058,216	99.7%
\$1,886,002	\$2,197,593	\$1,067,261	99.5%
\$1,955,330	\$2,295,369	\$1,076,505	99.1%
\$2,034,479	\$2,419,887	\$1,085,586	98.5%
\$2,101,170	\$2,512,895	\$1,094,697	97.7%
\$2,184,414	\$2,609,590	\$1,106,512	96.8%
\$2,266,765	\$2,732,078	\$1,118,188	95.6%
\$2,355,211	\$2,875,627	\$1,129,869	94.2%
\$2,450,235	\$3,001,751	\$1,143,795	92.6%
\$2,545,526	\$3,139,598	\$1,158,466	91.1%
\$2,626,644	\$3,287,948	\$1,174,080	89.4%
\$2,746,502	\$3,459,068	\$1,191,823	87.5%
\$2,847,923	\$3,626,103	\$1,209,916	85.6%
\$2,954,196	\$3,791,827	\$1,230,183	83.9%
\$3,069,057	\$3,996,988	\$1,253,668	82.2%
\$3,188,431	\$4,186,078	\$1,277,047	80.4%
\$3,291,581	\$4,363,866	\$1,302,269	78.6%
\$3,426,513	\$4,556,570	\$1,331,030	77.0%
\$3,563,555	\$4,773,183	\$1,360,412	75.1%



Portfolio value by age



Age

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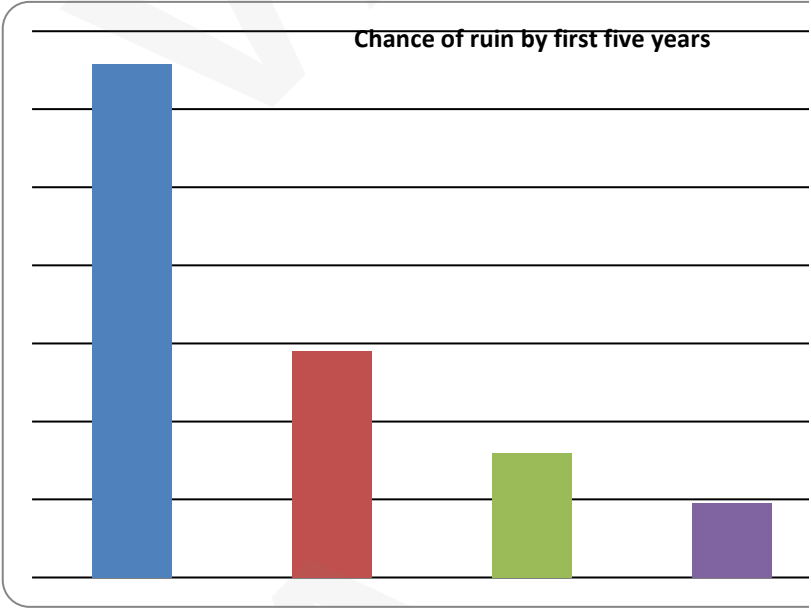
Ruin by the first five years of returns

Paths split into fifths by average real return in years one to five. Base case.

Fifth	Average yearly real return, years 1 to 5	Chance of ruin
Worst fifth	-3.8%	65.8%
Second	2.1%	29.1%
Middle	5.1%	15.9%
Fourth	7.9%	9.5%
Best fifth	11.9%	4.1%

Gap, worst to best fifth (points)	+61.7%
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Median ending wealth	
	\$0
	\$420,549
	\$922,220
	\$1,333,061
	\$2,163,346



This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across most of the width of the page. At the bottom center, there is a small, solid blue rectangle. The right edge of the paper is slightly curved, suggesting it might be a page from a notebook or a folder. There is no handwriting or other markings on the page.

Best stock weight under different assumptions

Lowest ruin across weights from 0% to 100%. Regimes and health shocks on.

Assumptions	Best static mix	Static ruin there	Best guardrail mix
Default	70%	24.8%	40%
Lower returns	80%	36.5%	50%
Normal returns model	70%	21.3%	40%
Spending \$50,000	100%	38.8%	70%
Retire at 60	80%	30.0%	50%

Ruin by stock weight

Stock weight	Default (static)	Default (guardrails)	Lower returns (static)
0%	62.4%	25.2%	79.5%
10%	48.6%	17.4%	69.0%
20%	37.5%	13.2%	58.0%
30%	30.8%	11.3%	49.0%
40%	27.6%	10.7%	43.1%
50%	25.7%	11.1%	39.7%
60%	24.9%	11.9%	37.9%
70%	24.8%	13.1%	36.9%
80%	25.1%	14.5%	36.5%
90%	25.9%	16.5%	36.7%
100%	27.0%	18.3%	37.3%

Guardrail ruin there	All bond ruin (static)	Guardrails beat static at every weight
10.7%	62.4%	Yes
20.0%	79.5%	Yes
6.4%	65.8%	Yes
24.4%	93.6%	Yes
16.2%	80.0%	Yes

Lower returns (guardrails)	Normal returns model (static)	Normal returns model (guardrails)
39.8%	65.8%	24.0%
30.5%	51.8%	14.6%
24.5%	39.1%	9.7%
21.4%	30.5%	7.2%
20.0%	25.4%	6.4%
20.0%	23.0%	6.5%
20.3%	21.8%	6.9%
21.4%	21.3%	7.7%
22.8%	21.5%	8.6%
24.4%	22.0%	9.9%
26.3%	22.7%	11.4%

Guardrail gain at best mixes (points)	
	-14.0%
	-16.6%
	-14.8%
	-14.5%
	-13.8%

Spending \$50,000 (static)	Spending \$50,000 (guardrail)	Retire at 60 (static)	Retire at 60 (guardrail)
93.6%	60.8%	80.0%	43.0%
86.9%	47.8%	66.8%	30.5%
75.6%	37.2%	53.1%	23.0%
63.6%	30.7%	42.9%	18.9%
54.4%	26.7%	36.5%	16.9%
47.8%	25.0%	32.9%	16.2%
43.7%	24.4%	31.1%	16.5%
41.3%	24.4%	30.2%	17.4%
39.7%	24.9%	30.0%	18.8%
38.9%	26.1%	30.4%	20.6%
38.8%	27.4%	31.3%	22.5%

Assumptions

Edit `retirement_sim/config.py` to change any of these. All money is in today's money.

Regime	Share of years
Expansion	72%
Bear	11%
Stagflation	16%
Long-run mix	

Item	Value
Starting portfolio	\$1,000,000
Ages	65 to 95
Spending	\$40,000 a year after tax
Base mix	60% stocks, 40% bonds
Regime memory	Markov chain, yearly
Long term care	From age 75
Acute health shock	3% a year
Health cost inflation	1.5% above inflation
Guardrails	10% cut, 10% raise
VPW	2.5% real return assumed
Tax	US single filer 2024
Forced withdrawals	From age 73
Paths	20,000
Ruin	Net cash short by more than 1% of the year's need

Stocks mean	
	11.0%
	-11.0%
	3.0%
	7.2%

Note
Taxable 300,000. Traditional 500,000. Roth 200,000.
Fixed horizon. Nobody dies early.
A 4% starting rate. Real terms.
Same mix in every account.
Expansion 89.5% stays. Bear 35% stays. Stagflation 75% stays.
1% a year rising to 5% at 95. \$60,000 a year for about 2.5 years. Once at most.
Lognormal, \$20,000 average.
Applied to all health costs.
Trigger at 20% drift in the withdrawal rate. Floor 70%, cap 150% of target.
Annuity due payout share on what is left after the health bill.
Brackets and deduction held in real terms. No state tax.
IRS uniform lifetime table, interpolated.
Random seed 42. Scenarios share the same draws.
Health bills are paid first, then spending.

Stocks volatility	Bonds mean	Bonds volatility	Stock and bond correlation
14.0%	2.5%	5.0%	0.10
24.0%	4.0%	8.0%	-0.20
19.0%	-2.0%	8.0%	0.60
17.8%	1.9%	6.3%	0.14